

UNIVERSITY OF HELSINKI

ENHANCING DESIGN THINKING THROUGH THE PRODUCT DESIGN WORKSHOPS

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DISSERTATIONES UNIVERSITATIS HELSINGIENSIS | 420 2025

Department of Education
University of Helsinki
Dissertationes Universitatis Helsingiensis 420/2025

Enhancing Design Thinking Through the Product Design Workshops

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ACADEMIC DISSERTATION

To be presented, with the permission of the Faculty of Educational Sciences of the University of Helsinki, for public examination via remote defence, on 21 November 2025 at 14:15.

Helsinki 2025

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Publisher: University of Helsinki
Series: Dissertationes Universitatis Helsingiensis 420/2025

ISBN 978-952-84-1568-8 (Print)
ISBN 978-952-84-1567-1 (PDF)
ISSN 2954-2898 (Print)
ISSN 2954-2952 (PDF)

PunaMusta, Joensuu 2025

ABSTRACT

Using design thinking (DT) skills in product design has become an essential component of design education. Various workshop settings offer opportunities for design students to develop expertise in design processes. This dissertation presents research that examines a product design-led and workshop-based pedagogical approach, regarding two workshop bases of the Cross-Disciplinary Workshop for Product Form Design (CDW for PFD, Studies I & II) and the Participatory Workshop for Culturally Oriented Product Design (PW for COPD, Study III). The aim is to investigate how students engage in product design and what design responses emerge when DT skills are taught and applied. The present study seeks to generate new insights into DT pedagogy by utilising two pedagogical workshops, each involving a distinct conceptual product design (PFD and COPD) within their respective settings (CDW and PW). It encompasses two cycles comprising three empirical studies based on educational design research (EDR) to analyse comparatively both the specific and inherent characteristics of DT. Holistically, data collection focused on students' visual representations, including textual, visual, and verbal artefacts. The qualitative analysis involved coding techniques across sub-studies, such as the analysis of design activities in PFD (Studies I & II) and metaphorical product design in COPD (Study III). This research specifically presents the findings from each study with respect to the characteristics of DT. The results revealed how the two workshop settings supported students' DT activities and generated five comparative pairs for discussion: the designer-as-user approach vs. direct user involvement, well-structured vs. open-ended tasks, talking sketches vs. thinking sketches, product form design vs. culturally oriented design, and collaborative group design vs. individual design. The results demonstrate that design educators can effectively support the development of DT skills by critically considering these paired contrasts in the design workshops. The research provides valuable theoretical contributions and practical implications for structuring a pedagogical design workshop that fosters DT capabilities, focusing on specific design activities in diverse, real-world contexts.

Keywords: design thinking; product form design (PFD); culturally oriented product design (COPD); educational design research (EDR); product design; product design workshop

TIIVISTELMÄ

Muotoiluajattelun taitojen hyödyntäminen tuotemuotoilussa on muodostunut olennaiseksi osaksi muotoilukoulutusta. Erilaiset työpajaympäristöt tarjoavat muotoiluopiskelijoille mahdollisuuksia kehittää osaamistaan suunnitteluprosesseissa. Tämä väitöskirja tarkastelee tuotemuotoiluun ja työpajapohjaiseen pedagogiikkaan liittyvää lähestymistapaa, johon sisältyvät muodon suunnittelun poikkitieteellinen työpaja (tutkimukset I & II) sekä kulttuurisesti suuntautuneen tuotesuunnittelun osallistava työpaja (tutkimus III). Tavoitteena on selvittää, miten opiskelijat osallistuvat tuotemuotoiluun ja millaisia suunnitelmia syntyy, kun muotoiluajattelun taitoja opetetaan ja sovelletaan. Tutkimus tuottaa uusia näkökulmia muotoiluajattelun pedagogiikkaan hyödyntämällä kahta pedagogista työpajaa, joista kumpikin keskittyy erilaiseen käsitteelliseen tuotemuotoiluun omissa ympäristöissään. Työ käsittää kaksi sykliä ja kolme empiiristä tutkimusta kasvatustieteellisen suunnittelututkimuksen viitekehyksessä, jotta muotoiluajattelun erityisiä ja luontaisia piirteitä voidaan vertailevasti analysoida. Aineistonkeruu painottui kokonaisvaltaisesti opiskelijoiden visuaalisiin esityksiin – tekstuaalisiin, visuaalisiin ja verbaalisiin artefakteihin. Laadullisessa analyysissä hyödynnettiin erilaisia koodaustekniikoita muodon suunnittelutoiminnan analyysissä (tutkimukset I & II) ja metaforisen tuotemuotoilun analyysissä kulttuurisesti suuntautuneen tuotemuotoilun kontekstissa (tutkimus III). Tutkimus esittelee kunkin osatutkimuksen tuloksen muotoiluajattelun ominaispiirteiden valossa. Tulokset osoittavat, kuinka kaksi työpajaympäristöä tukivat opiskelijoiden muotoiluajattelua ja tuottivat keskusteltavaksi viisi vertailuparia: suunnittelija käyttäjänä vs. suora käyttäjäosallistuminen, hyvin jäsennelty tehtävä vs. avoin tutkiminen, keskusteluluonnokset vs. ajatteluluonnokset, muodon suunnittelu vs. kulttuurisesti suuntautunut suunnittelu, yhteistoiminnallinen suunnittelu vs. yksilöllinen osallistava lähestymistapa. Tulokset osoittavat, että muotoilun opettajat voivat tehokkaasti tukea muotoiluajattelun taitojen kehitystä tarkastelemalla kriittisesti näitä vastinpareja työpajoissaan. Tutkimuksen johtopäätökset tarjoavat arvokasta ohjausta pedagogisen suunnittelutyöpajan rakentamiseen muotoiluajattelun valmiuksia vahvistavalla tavalla, painottaen suunnittelutoimintaa moninaisissa, todellisen elämän konteksteissa.

Avainsanat: muotoiluajattelu; muodon suunnittelu; kulttuurisesti suuntautunut tuotemuotoilu; kasvatustieteellinen suunnittelututkimus; tuotemuotoilu; tuotemuotoilun työpaja

ACKNOWLEDGEMENTS

First and foremost, I want to thank my supervisors, Pirita Seitamaa-Hakkarainen, Henna Lahiti, and Seija Kairavuori. Your guidance and openness to new ideas have been invaluable throughout this doctoral journey. I express my deepest respect and gratitude to Pirita Seitamaa-Hakkarainen for her sustained research support. Working with her at the University of Helsinki and engaging in discussions on various research-related issues enhanced my understanding of research. Even after I returned to China, our collaboration continued online, which has been one of the more meaningful aspects of this journey. My second supervisor, Henna Lahti, has consistently provided professional theoretical and methodological guidance that has been indispensable to me in undertaking this doctoral research. Her insights have deepened, refined, and advanced the research process. Her unwavering support enabled me to accomplish thesis writing that I had not previously thought possible, and I am grateful for her contributions. I also extend my sincere thanks to my third supervisor, Seija Kairavuori, for her thoughtful review and valuable contributions, which have significantly benefited this study.

Furthermore, I want to express my sincere thanks to Sirpa Kokko and Mirja Kälviäinen, who served as the pre-examiners for this dissertation and provided insightful comments and suggestions that strengthened the research.

This research primarily focuses on university product design education in China and addresses complex and multifaceted issues. The study involved numerous product design professionals and encountered challenging problems throughout the doctoral research process. As the research progressed, new issues emerged and became involved. Through continuous exploration and collaboration with three supervisors, the research co-evolved, enabling greater depth and clarity. This process helped me to understand these challenges more clearly, and I am especially thankful for their support, which made this meaningful transformative possible.

Special thanks are due to the University of Helsinki (Finland) and Yunnan University of Finance and Economics (YUFE) (China). The University of Helsinki has provided valuable doctoral courses that offered diverse perspectives and methodological insights. I also offer sincere thanks to the leadership of the School of Communication and Design Art (SCDA) at YUFE for their support. Their encouragement and facilitation were instrumental in the successful completion of this doctoral research.

Further, as the head of SDRC-YUFE, the Sustainable Design Research Centre (www.linkedin.com/in/weishu-yang-b56a61307), I have integrated design pedagogy and research work grounded on the scholarly research programmes of YUFE, across multiple lines, thereby driving co-innovation among design projects and expecting to achieve long-

term, commercially sustainable innovation. SDRC-YUFE is a research centre grounded in design studies and team-based approaches, building transitions from the micro level (advances in on-site and metaverse-based service design technologies) to social network development and to macro-level external environments, which helps organise theoretical frameworks and methodological practices. My deepest gratitude goes to my colleagues, Dr Kun Wang, Dr Hang Shang, and Bochen Guo, for their continuous support.

This research was funded by the scholarly research programmes of Yunnan University of Finance and Economics:

- Research on the Pedagogy of Cultural and Creative Product Design Based on the Beautiful Villages Construction: A Case Study of Masa Village, Maguan County (grant number 2020C01)
- 2024 First-Class Curriculum Construction Project: Product Design Survey and Research Method, the School of Communication and Design Art
- Service Design Course Pedagogy Reform Based on Digital Ethnography Methods and Metaverse Application Technology. Notice on the Publication of the List of 2025 Undergraduate Education and Teaching Reform Research Projects (University Level), the Office of Academic Affairs, Document No. (2025)170

Further support was provided by SCDA-YUFE, China.

Yunnan Kunming, SDRC-YUFE, 7/10/2025

Yang, Weishu 杨维抒

List of original publications

This dissertation is based on the following publications, which are referred to in the text by their corresponding Roman numerals (I, II, and III).

I Yang, W., Lahti, H., & Seitamaa-Hakkarainen, P. (2022). Creating new 3D forms in collaborative product design. *Design and Technology Education: An International Journal*, *27*(1), 9–28. Retrieved from https://openjournals.ljmu.ac.uk/DesignTechnologyEducation/article/view/1149

II Yang, W., Lahti, H., & Seitamaa-Hakkarainen, P. (2022). A Cross-Disciplinary Workshop on 3D Modelling: Students' Design Concepts in the Bottle Form Design Task. *Journal of Design Thinking*, *3*(2), 201-216 doi: 10.22059/jdt.2023.357801.1094

III Yang, W., Lahti, H., & Seitamaa-Hakkarainen, P. (2025). Three levels in culturally oriented product design: a participatory approach to cultural inspiration in design education. *Design and Technology Education: An International Journal*, *30*(1), 63–85. Retrieved from https://openjournals.ljmu.ac.uk/DesignTechnologyEducation/article/view/3043

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1 INTRODUCTION

Creativity is well documented in design-related fields as an essential aspect of the human mind and activity, playing a significant role in culture and history (Sawyer 2012; 2018). It plays a critical role in today's innovation-driven knowledge society, in which digitalisation increasingly automates routine and cognitively skilled work. Knowledge workers are now expected to engage in design activities that add value to knowledge production and product development. This shift necessitates a stronger commitment to creativity and design education to prepare innovative professionals for rapidly evolving industries. The growing demand for innovation has also prompted a restructuring of design education to promote cross-disciplinary and multidisciplinary teaching, learning, and research (Zhang et al., 2015). In recent decades, using design thinking (DT) skills in product design has become an essential component of design education (Koh et al., 2015).

In today's design field, cross-disciplinary design refers to a creative and educational model that intentionally integrates methods, tools, and perspectives from multiple traditionally distinct design fields, such as product, textile, graphic, and interior design, alongside non-design domains like business and technology. This approach has evolved in response to the increasing complexity of contemporary industry challenges that designers face, requiring collaboration that transcends disciplinary boundaries (Rodgers & Bremner, 2019; Zhang et al., 2015). In professional design practice, the shift towards project-based and fluid collaboration has become a defining feature of the design industry. Rather than relying on solid expertise, cross-disciplinary project work emphasises the co-creation of solutions through diverse stakeholder involvement and the shared use of advanced digital tools and production technologies (Rodgers & Bremner, 2019). Zhang et al. (2015) suggest that designers today must possess not only core design skills, but also the ability to understand and address multiple stakeholders within the broader context of business (Rodgers & Bremner, 2019; Kim & Yoon, 2012) and systematic design integrations (Ekmekci & İstanbul, 2019; Zhang, Chai, & Tan, 2003; Zhang, Tan & Chai, 2003; Yang, 2024a; 2025).

This paradigm shift is increasingly reflected in design education, for which cross-disciplinary pedagogy plays a critical role in supporting design educators. Institutions such as Stanford's d.school (2010), the mission of which is to advance design usage as a tool so that design practitioners can strengthen their creative capacity with the goal of leaving the world better than they found it, and Aalto University's Design Factory (Björklund et al., 2013; Lou et al., 2016) exemplify how educational environments can foster cross-disciplinary collaboration by blending design with business, technology, and engineering. Some empirical studies confirm the value of such configurations with global product-

design-engineering programmers (De Vere et al., 2010), cross-disciplinary communication, and co-creation design (Fleischmann, 2022). In China, the growing investment in cross-disciplinary design education further reinforces this trend (Zhang et al., 2015; Herr, 2018; Zhao, 2023). For design educators, cross-disciplinary design pedagogy offers effective strategies to develop students' innovation capabilities and prepare their skills for challenging real-world complexity. A cross-disciplinary design approach could demonstrate how such a pedagogical experimental model supports the cultivation of T-shaped designers (Nae, 2017), those with deep expertise in a specific area and broad collaborative skills across other domains, particularly in the design field. According to Nae (2017), three educational models illustrate this cultivated approach: 1. a foundation curriculum that exposes students to a wide range of design subjects; 2. targeted opportunities to deepen secondary areas of expertise, facilitating knowledge cross-over and the development of expert generalists; and 3. programmes focused on expanding horizontal skills to meet industry demands for team-oriented and collaborative practices. The cross-disciplinary learning approach enables design students from a range of design majors to address the product design domain practically and broaden their expertise. In this dissertation, the term "cross-disciplinary workshop" refers to a product design course in which students from various design disciplines engage in the same lectures and practical sessions. During the collaborative design sessions, students work alongside their peers from their respective design disciplines.

Additionally, the participatory design approach for learning product design offers a novel learning mindset that fosters deeper creative capabilities in real-world problem-solving contexts. In particular, within the field of cultural product design, authentic learning environments provide cultural contexts for design practice. Through workshops, student designers and design educators can gain a greater appreciation of the impact of integrating cultural inspiration into cultural product innovation by developing creative ideas and solutions (Moalosi et al., 2010; Qin & Ng 2020), as well as culturally sustainable design research (Yang, 2024b). This design approach benefits both practice and education, as it helps design educators to understand how to incorporate cultural elements into product design, and use cultural sources as a pedagogical set. The culturally oriented product design approach could exemplify this integration through a participatory approach within the local community. This participatory influence, echoing Manzini (2016), underscores the importance of democratic engagement in design; here participatory practices help uncover and respond to authentic community needs. He contrasts "solution-ism" with "participation-ism," arguing that meaningful participation arises when facilitators avoid imposing expert perspectives and instead foster equitable knowledge exchange with communities (Manzini, 2016).

The present study investigates the inherent characteristics of DT practices observed when design students engage in product design in design practices. The aim of the dissertation is to understand the nature of students' DT practices in two product design workshops, and to enhance pedagogy in product design through empirical examination. The main research questions are as follows:

RQ1: What was the nature of students' design thinking practices in the Cross-Disciplinary Workshop for Product Form Design (CDW for PFD) and the Participatory Workshop for Culturally Oriented Product Design (PW for COPD)?

RQ2: How is learning in DT enhanced in the CDW for PFD and the PW for COPD?

This present study consists of two cycles, comprising three design workshop–based empirical studies. The empirical studies were structured around two design workshops to enable comparison with DT characteristics. The first cycle, Studies I and II, was based on the cross-disciplinary CDW for PFD and conducted on campus, using rapid modelling techniques for bottle form design, i.e., this design task is developing the form of a detergent bottle (product) to meet client's requirements, to facilitate product formation. The second cycle, Study III, was based on the participatory PW for COPD and the aim was to facilitate culturally oriented product design in an authentic off-campus field study. RQ1 is addressed in the results section, and RQ2 in the discussion section.

The on-campus workshop focused on product form and functional features, whereas the off-campus workshop emphasised cultural understanding, content, and design implications. These distinct design tasks were deliberately chosen and jointly presented through a situated learning approach, integrating students' learning objectives with the research objectives in both on-campus courses and fieldwork. The contrast between product-form design and culturally oriented design establishes the disciplinary foundation for two routine design practice activities: one situated within the on-campus courses, and the other in the field environment. Accordingly, the design tasks were embedded in the regular activities of two required design workshops to support the related selective courses while also forming part of the specialised curriculum to be studied and further developed within the product design discipline.

This research demonstrates the pedagogical value of engaging design students from diverse design backgrounds in real-world design contexts. Using a product design-led and workshop-based approach positions DT as a key competency for fostering innovation among student designers. The study is grounded in university product design pedagogy, with the educational content specifically developed and implemented for this purpose. Based on this framework, empirical studies were applied at Yunnan University of Finance and Economics (YUFE), China. The results highlight the effectiveness of design workshop approaches and support the continued development and adoption of the CDW and PW frameworks in product design education. The research provides valuable theoretical contributions and practical implications for a university-level design pedagogy of product design-led design workshops.

2 THEORETICAL BACKGROUND

This chapter examines the concept and discourse surrounding design thinking. First, two key strands of discourse are identified in design research: one centred on "designerly thinking," rooted in traditional design research, and the other on "design thinking" as applied in business and organisational contexts (Kimbell, 2011; Johansson-Sköldberg et al., 2013). The enactment of DT is described through five core characteristics: user focus, problem framing, visualisation, experimentation, and diversity (Carlgren et al., 2016). Second, DT in product design is examined in more detail.

2.1 Design thinking definition and enactment

DT describes both how designers work and the design processes they use to solve problems. At its core, DT focuses on solving challenges creatively with user involvement (Cross, 2001; 2023; Hay et al., 2020). Over time, its meaning expanded from a design method to a broader approach used in business, social innovation, and management (Buchanan, 1992; Cross, 2023; Kimbell, 2011; 2012; Johansson-Sköldberg et al., 2013). The shift also helped tackle "wicked problems," or complex design challenges. The term design thinking first appeared in the 1960s, when design research became a formal academic field. Scholars like Cross (1984; 2023), and Margolin and Buchanan (1995) studied how designers framed problems, generated ideas, and developed solutions through iteration. Early design thinking emphasised structured problem-solving methods within professional settings.

Peter Rowe (1987) formally introduced the term in his book Design Thinking, describing it as the unique cognitive approach designers use, and emphasising how designers frame problems and explore solutions creatively. Around the same time, related research by Cross (1984; 2001; 2006) highlighted how designers think differently compared to those in other fields and extended this into the idea of a "designerly way of knowing," focusing on designers' special thinking patterns and decision making. This academic view stressed expertise, creativity, and reflection-in-action, as Schön (1983) described, where designers adjust their actions based on real-time feedback.

By the late 20th and early 21st centuries, the meaning of DT changed significantly. Cross (2023) noted the difference between original "design thinking" (for professional designers) and the newer "design thinking" (DT) used in business. Kimbell (2011) similarly points out that although the term started in academia, it is now used mainly in business. In the business field, DT became a tool for innovation, helping solve problems creatively with a user-centred focus (Brown, 2008; 2009). DT fosters innovation by focusing on user needs and producing emotionally satisfying, practical solutions. Influential leaders like Tim

Brown from IDEO, and Roger Martin, who promoted DT to boost business creativity and competitiveness. IDEO is an American design innovation consultancy known for pioneering DT, with the approach using human-centred design, experimentation, collaboration, and prototyping to solve complex problems. In these settings, DT is often simplified so that non-designers can use it to solve organisational challenges (Liedtka, 2015). DT typically follows a structured process, helping teams creatively solve complex problem challenges. A well-known model from Stanford's d.school includes five phases: Empathise, Define, Ideate, Prototype, and Test (Stanford d.school, 2010). The Hasso Plattner Institute in Germany has further developed this model by breaking down the empathise phase into two distinct components: understand and observe. This adaptation emphasises a user-centred design (UCD) approach to defining design challenges and seeks solutions through an iterative, non-linear process. However, evaluating DT is still challenging. Traditional evaluation methods do not easily capture its iterative, user-focused processes. Carlgren et al. (2016) emphasised that DT is enacted through five core characteristics: user focus, problem framing, visualisation, experimentation, and diversity, each of which challenges standardised forms of evaluation and calls for context-sensitive, process-oriented approaches. Table 1 shows the DT framework (Carlgren et al., 2016) for design enactment adopted for the present study.

Table 1 The DT framework adapted from Carlgren et al. (2016)

Characteristic	Practical explanation	Essential features
User focus	Seek to understand the latent needs and pain points of users (empathise), and let this understanding guide conceptual design, and involve intended users in ideation and visualisation.	Pose assumptions based on personal experience to empathise with the people who are the intended users of the context; Structuring informal meetings with the stockholders, such as clients and users.
Problem framing	Implemental practice activities of DT within the cognitive process. Reframe the initial problem to expand both problem and solution space for satisfying solutions.	Iteratively framing and reframing design problems by talking, writing, and gathering information to explore solutions.
Visualisation	Make ideas and insights visual and tangible by externalising knowledge and creating new ideas. The ideas are communicated by visual means.	Creation of visual representations through ideating and exploring visual features based on the produced solutions.
Experimentation	For promoting participants' DT capability of engaging in design processes by the designed tasks structuring design practice iteratively.	Creation of flexible and physical space that supports experimentation and visualisation, including structuring experimental conditions in design workshop.
Diversity	Create diverse groups and let everyone's opinion count. Collaborate with external entities to seek diverse perspectives and inspirations (variety of fields, broad research).	Structuring active participation for engaging DT in various attractions and events offers experiences and ideation activities.

The five focus characteristics identified in Table 1 illustrate the multifaceted nature of DT in a design process. To engage in the workshops successfully, designers need support in 1. recognising user perspectives to frame design problems; 2. understanding the DT problem rationale guiding design practice; 3. effectively using visualisation in design processes; 4. structuring design practice iteratively; and 5. reflecting on and sharing their emerging design knowledge with various participants. Being aware of these thematic foci may help to harness the complexity of the design process.

2.2 Design thinking in product design

Integrating user perspectives is a core principle of DT practice. Through user participation (direct or indirect), design outcomes could become more desirable and inclusive. DT also refers to the problem-solving processes intrinsic to design practice (Schön, 1983; Cross, 2001; Hay et al., 2020; De Vere et al., 2010; Fleischmann, 2022). This chapter introduces two theoretical approaches to design problem-solving, Simon's (1969) rational problem-solving model and Schön's (1983) reflection-in-action model. The concept of problem–solution co-evolution is also addressed, which emphasises that design processes are collaborative and iterative rather than linear; understanding of the problem and development of solutions evolve simultaneously. There is also an examination of visual tools such as sketching, 3D modelling, and mind mapping. Lastly, this chapter introduces product concept design as an integral phase of DT practice.

2.2.1 User focus

Understanding and integrating the user perspective has become a foundational principle in DT practice, ensuring that design outcomes are more desirable, culturally sensitive, and impactful (Sanders & Stappers, 2008; Liedtka, 2015). Interdisciplinary training environments, such as global product design engineering programmes and mixed-discipline design workshops, have been shown to broaden students' appreciation of diverse user groups and market contexts, thereby sharpening empathy driven design skills (De Vere et al., 2010; Fleischmann, 2022). As Liedtka (2015) notes, modern DT frameworks emphasise user-centred and user-driven approaches as being essential to effective design. Whether through direct participation or indirect involvement, the incorporation of user insights (emotional, cultural, and social) is crucial for creating meaningful and relevant solutions in the design process (Brown, 2008; Sanders & Stappers, 2008).

In design practice, user involvement can take two primary forms, direct and indirect (Zhang & Dong, 2016; Moghimi et al., 2017). Direct participation sees users actively engaged in the design process, contributing ideas, feedback, and experiential knowledge. Here, users often assume a co-designer role, shaping solutions collaboratively with designers (Zhang & Dong, 2016). In contrast, indirect user involvement relies on secondary data sources such as literature reviews, case studies, and domain expertise to interpret user needs without their physical participation (Moghimi et al., 2017). For instance, Zhang and Dong (2016) identified four models for representing user perspectives, including the

“Designers Representing the Users” model, through which designers synthesise user needs through research and interpretation. Similarly, Moghimi et al. (2017) emphasised incorporating user values into housing design through indirect methods, highlighting that a deep understanding of user priorities can significantly enhance design quality even without direct user engagement.

Overall, whether through direct collaboration or indirectly interpreting secondary data, embedding user perspectives in the design process remains a critical strategy in achieving human-centred and contextually inclusive design outcomes. Both methods underline the essential role of empathy, cultural understanding, and feedback in design practices (Sanders & Stappers, 2014). Typically, the UCD study (Abras et al., 2004) emphasises the iterative nature of designing, involving the idea that usability and user experience remain central throughout the design lifecycle. It also highlights design methods for user research and evaluation, as well as the benefits of aligning systematic design with users’ needs, tasks, and contexts. Applying user perspectives and participatory design approaches in an experimental study to commercial product development (Wilkinson & De Angeli, 2014), researchers investigated how participants interacted within a simulated shopping environment. These activities were examined through video recordings, talk-aloud protocols, and questionnaires. My previous study presented a product-service systems design grounded in the research results, in which designers used market survey as a data collection and analysis means for the subsequent design phase (Yang, 2024a). The market survey collected the data, and analysis generated the research results. The study by editors Keinonen and Takala (2006) emphasises integrating UCD into product concept design, stressing that user insights influence not only product usability but also innovation, ensuring that products are both functional and resonate with users.

2.2.2 Problem-solving and exploring solutions

In design practice, the processes of problem-solving and solution exploration are shaped by two foundational yet contrasting perspectives from the renowned scholars Simon and Schön in cognitive science: rational problem-solving and reflection-in-action. These perspectives serve as theoretical cornerstones for understanding how designers approach complex contextual challenges and generate innovative outcomes.

The rational problem-solving model, proposed by Simon (1969), frames design as a systematic, goal-oriented activity grounded in bounded rationality. Problems are treated as being clearly defined and solvable through structured analysis and iterative refinement. This model, deeply rooted in early cognitive science and artificial intelligence (Newell & Simon, 1972), emphasises analytical decomposition, rule-based reasoning, and optimisation within predefined constraints. According to this view, designers follow linear processes to identify objectives, explore alternatives, and converge on optimal solutions.

In contrast, Schön’s (1983) concept of reflection-in-action presents design as a situated and interpretive practice. Here, design problems are seen as ill-structured, ambiguous, and evolving, shaped through the designer's engagement with materials, users, and context.

Rather than applying fixed knowledge, designers engage in reflective conversations with the situation, responding to unexpected conditions with intuition, improvisation, and iterative reframing (Schön, 1983; Cross, 1984; Dorst & Dijkhuis, 1995; Visser, 2006). This view foregrounds the co-evolution of problem and solution, and positions design as an emergent dialogue adaptive process.

A key contribution of Schön's approach lies in highlighting that design problems are fluid and socially constructed. As designers interact with stakeholders and confront new constraints, their understanding of the problem space is transformed. Group reflection and shared framing problems, especially in collaborative and participatory contexts, play a crucial role in navigating uncertainty and aligning multiple perspectives (Tan et al., 2023). The reflective practitioner model has evolved to emphasise its relevance in team-based and socially oriented design practices, through which reflection enables meaningful engagement and ethical responsiveness during challenging complex contexts. Visual tools such as sketching, modelling, and prototyping act as enablers of reflection and solution exploration. They provide a means for externalising thought and engaging in an ongoing "visual dialogue" with the developing artefact (Crilly, 2021). This process supports both individual and collective sense-making and problem-reframing.

In educational contexts, the way design problems are framed also significantly influences students' creative engagement. As Kapkın and Joines (2021) argue, the design brief, and more specifically the problem statement, can serve as a creativity catalyst by priming students' cognitive engagement and expanding the perceived problem space. Effective problem framing not only guides solution generation but also opens up opportunities for innovation by encouraging divergent thinking early in the process.

Importantly, these two paradigms, rational problem-solving and reflection in action practice, should not be seen as mutually exclusive. Schaathun (2022) has said that Simon and Schön, though distinct in emphasis, share common ground in acknowledging the need for rationality within constraints. Both recognise that designers operate in uncertain environments and must navigate between structure and flexibility. Simon stresses structured design reasoning and goal-oriented design, while Schön underscores situated adaptation and the creative negotiation of meaning.

Integrating both perspectives provides a holistic view of design. It allows educators, and researchers to appreciate the dynamic interplay between problem understanding and solution creation, logic and intuition, structure, and emergence. Ultimately, exploring solutions in design is not merely about selecting from known options, but about iterative problem-framing and exploring solutions, generating new interpretations, and engaging with complex material and social responses.

2.2.3 Problem-solution co-evolution models

The problem–solution co-evolution model is an important framework in design research that explains how design processes are collaboratively iterative, not linear. It shows that both the understanding of the problem and the development of solutions co-evolve over time

(Maher & Poon, 1996; Maher & Tang, 2003; Dorst & Cross, 2001; Lonchampt et al., 2004). Maher and colleagues (Maher & Poon, 1996; Maher & Tang, 2003) introduced this idea, emphasising that design problems are ill-structured, meaning that they rarely present complete requirements from the start. Instead, requirements and constraints shift and grow as designers gain new insights and refine both the problem and potential solutions. Maher and Poon (1996) proposed an algorithmic model in which understanding the problem and generating a solution were developed in tandem. Their diagrammatic representation became widely recognised for capturing the dynamic, evolving nature of real-world design activities.

Dorst and Cross's (2001) main insight was that creativity in design comes from simultaneously developing both problems and solutions. Designers often shift between problem understanding and solution exploration, leading to innovative breakthroughs (Maher & Tang, 2003). Creativity, therefore, is not just about solving a predefined problem, but is also about discovering and framing new problems during the design process.

The co-evolution model is especially relevant in collaborative design when teams of designers or stakeholders work together. Lonchampt et al. (2004) applied the model to group design and identified four main activities:

1. Conjecture: proposing early ideas,
2. Definition: clarifying the problem,
3. Evaluation: assessing both problem framing and solutions,
4. Reformulation: updating the problem and refining solutions based on feedback.

These activities unfold in shared spaces, enabling team members to negotiate and reflect collaboratively. Through this iterative interaction, teams co-construct shared understanding and direction. The study also emphasised the roles of appropriation and emergence. As designers' activities move between problem and solution spaces, new ideas and criteria emerge. These transitions drive innovation and creativity, showing that designers not only respond to constraints but also generate new design opportunities through reflective iterative activities.

Recent research has further expanded and formalised the co-evolution model. Gero et al. (2022) provided a more abstract and formal structure of the model, explaining the transitions between problem and solution states. Cash et al. (2023) offered empirical studies that support the cognitive co-evolution process, showing how it explains the real ways designers think and act in design practice.

In summary, the co-evolution model captures how designers continuously adapt, reframe problems, explore alternatives, and refine both their goals and solutions. It mirrors the dynamic realities of professional design work and underlines the importance of adaptability, exploration, and reflection. The model provides valuable insights into how design cognition operates, especially in complex, ill-structured, and collaborative design settings.

2.2.4 Visualisation

Visual representations are essential in design practice, helping designers develop, explore, and communicate their ideas. These can include initial sketches, 3D models, and detailed diagrams. According to previous research (Visser, 2011; Bertschi et al., 2012), visualisations allow designers to express abstract ideas clearly and support iterative development from early concepts to final details. These visuals often combine text, images, and symbols to organise complex information in an accessible way. Further, visualisation maps concepts graphically, making them easier to understand, especially when designers need to share ideas with teams or external stakeholders.

Throughout the design process, visualisation tools help organise thoughts and guide decision-making (Ferguson, 1992; Menezes & Lawson, 2006; Lotz & Sharp, 2017; Bresciani, 2019). Bresciani (2019) notes that both textual and visual elements like lines, shapes, and icons are used to structure designers' thinking. Among visual tools, sketching is especially important. Sketches allow designers to explore ideas quickly, rethink them, and move towards better solutions. Menezes and Lawson (2006) highlight how the cycle of sketching, reinterpreting, and evaluating drives creative idea generation. Sketches are typically rough and open-ended, which makes them ideal for early-stage exploration. Ferguson (1992) identified three main types of sketches: thinking sketches, talking sketches, and prescriptive sketches. In group settings, sketching supports dialogue and shared understanding. Designers often engage in co-sketching, drawing while discussing, to develop ideas together (Lotz & Sharp, 2017). These collaborative sketches help evolve design concepts through continuous feedback.

As design advances, 3D modelling and computer-aided design (CAD) tools become more important, particularly in product and architectural design (Krish, 2011; Bernal et al., 2015; Alcaide-Marzal, 2020). Krish (2011) points out that CAD systems help designers create models that combine geometric, functional, and manufacturing information, enhancing both creativity and technical accuracy. Generative 3D modelling, through which algorithms generate design variations, further expands creative options (Alcaide-Marzal, 2020). These show that CAD modelling encourages the discovery of new ideas.

Mind mapping is another valuable visual tool, especially in early-stage concept development. Mind maps help designers organise ideas and see relationships between distinct parts of a problem. Dong et al. (2021) argue that mind mapping encourages divergent thinking, crucial for the ideation phase. Their study found that students who used mind maps were better at analysing topics and integrating insights into their final designs. Combining mind mapping with sketching enhances analysis depth and strengthens the connection between different design elements, supporting both creative exploration and practical refinement.

2.2.5 Product concept design and its practice

Product concept design serves as the foundational phase in the broader product design process for new product development, closely tied to manufacturing and commercial viability (Keinonen & Takala, 2006; Ralph & Wand, 2009; Wölfel et al., 2013; Andreasen et al., 2015). It begins by identifying user needs and business goals and evolves into the conceptualisation of solutions that balance innovation with technical feasibility (Imbesi & Scataglini, 2021; Dahiya & Kumar, 2021; Keinonen & Takala, 2006). While the objective is to develop refined specifications for production, product concept design emphasises the exploration of ideas that satisfy both customer expectations and production realities.

Product concept design represents a critical phase of DT, serving as an integral step in conceptual product development. In this phase, DT is recognised as a broader problem-solving and innovation methodology that spans multiple stages, commonly aligned with the Double Diamond also called DT models (Design Council, 2015). With concept design occurring primarily during the ideation, it follows an iterative process of exploring many ideas (divergence) and narrowing them down to the ideal solutions (convergence) (Andreasen et al., 2015). During this stage, designers translate abstract user and contextual needs into tangible propositions by producing technical drawings, digital models, or physical and detailed components. These outputs define the intended shape, function, and assembly structure logic, thus informing downstream engineering and manufacturing processes. Importantly, this stage shapes not only the physical product but also the shared understanding of the design direction among team members and stakeholders.

From a DT perspective, the conceptual design phase occupies the “define” and “ideate” stages, whereby ideation is blended with structured decision-making (Andreasen et al., 2015). According to their study, this phase involves defining functional structures, specifying solution principles, and visualising configurations in ways that are not yet constrained by materials or production methods, but instead, explore what is functionally, creatively, and strategically possible. It is inherently an abductive reasoning process, wherein divergent and convergent thinking are used iteratively to explore, evaluate, and refine alternative solutions.

Conceptual design involves a cognitive process that integrates multiple modes of reasoning, such as analytical, visual, and intuitive thinking (Keinonen & Takala, 2006; Wölfel et al., 2013; Andreasen et al., 2015). Wölfel et al. (2013) argue that conceptual design in industrial design settings is a knowledge-driven activity wherein knowing, reasoning, and visualising operate simultaneously. Through sketching, modelling, and diagramming, designers develop both internal and shared understandings of form, function, and interaction. These external representations enable the exploration of ambiguity and uncertainty, either collaboratively or individually, which is particularly effective in workshop-based settings, where iterative feedback and prototyping are central.

In product-design, workshop-based design practice positions conceptual design as the basis of conceptual innovation in design, allowing designers to navigate complexity through iterative exploration and critical reflection (Sawyer, 2012; 2018; Andreasen et al., 2015). In

such a design setting, conceptual design becomes the operative mode of inquiry, structuring how form, function, and meaning are envisioned, ideated, articulated, and evaluated before physical prototyping begins. Grounded in product design thinking and structured through workshop formats, this approach offers distinct, yet complementary pathways based on product concept design practices such as designing product form, or cultural meaning.

In product form conceptual design, workshop-based practices focus on cultivating students' ability to navigate the visual, tactile, and semantic dimensions of product development. According to Keinonen and Takala (2006), product concepts must address more than technical feasibility; they should also express symbolic and aesthetic values, form language, proportion, material tactility, and emotional aspects. This multifaceted approach is deepened when formal product characteristics are explicitly tied to intended use contexts and user experiences, as emphasised by Andreasen et al. (2015). Design workshops structured around form, such as prototyping design techniques, create a platform for students to iterate through 2D sketches, mock-ups, and 3D models, developing their visual language while critically evaluating usability and emotional response. According to Crilly et al. (2009), design intentions related to user experience include drawing attention to the product; fostering recognition of product type; generating attraction; supporting comprehension of function; encouraging attribution of qualities; promoting personal identification; stimulating emotion; and provoking action. These intentions are shaped through external representations, sketches, CAD design, and physical models that act not only as communication devices, but also as cognitive tools for reasoning (Visser, 2011; Wölfel et al., 2013). Within the workshop environment, tool applications support design feedback loops and reflective practice, enabling design practitioners to align design ideas with both user expectations and production requirements.

On the other hand, culturally oriented conceptual design requires designers to embed cultural metaphors, symbols, and narratives into product concepts, making it an inherently interpretive and context-sensitive process. Keinonen and Takala (2006) emphasise that conceptual design must respond to its socio-cultural context, and in this case, that context includes local heritage, community identity, and intangible cultural values. Within workshop environments, this kind of design work is supported by participatory and field study methods, which allow student designers to investigate and engage with cultural contexts. Workshops for culturally oriented design use tools such as storytelling, metaphor design, and cultural meaning mapping to guide students through the process of translating abstract cultural ideas into tangible product features (Qin & Ng, 2020). These conceptual tools support semiotic encoding the transformation of beliefs, behaviours, and social meanings into product attributes, symbols, materials, and interactions (Qin & Ng, 2020). When employed within a structured framework, such as the three-phase model of identification, translation, and implementation (Dong et al., 2023), the design process enables more profound engagement with cultural narratives and results in culturally sensitive design outcomes.

Workshop-based learning provides a powerful structure for applying and developing conceptual design thinking. Andreasen et al. (2015) highlight the approach importance of

structured DT formulation in the conceptual design phase, enabling student designers to confront ambiguity, manage complexity, and generate solutions that are both imaginative and grounded. By iteratively exploring design proposals through drawing, modelling, prototyping, and reflection, designers actively construct meaning and evolve design intentions in response to feedback and constraints.

3 RESEARCH QUESTIONS

In this present study, I investigated the inherent characteristics of DT activities observed when design students engage in the product design processes. The aim of the dissertation is to understand the nature of students' DT practices in product-design workshops, and to enhance pedagogy in product design through empirical study. In particular, the aim of the empirical study was to examine product form design and culturally oriented product design in the design workshop settings for DT pedagogy. The main research questions are as follows:

> **RQ1**: What was the nature of students' design thinking practices in the Cross-Disciplinary Workshop for Product Form Design (CDW for PFD) and the Participatory Workshop for Culturally Oriented Product Design (PW for COPD)?
>
> **RQ2**: How is learning in DT enhanced in the CDW for PFD and the PW for COPD?

The present study consists of two cycles, comprising three design workshop–based empirical studies. The purpose is to generate new insights into DT pedagogy by utilising two pedagogical workshop configurations involving distinct conceptual product design goals. The empirical studies were structured around two design workshops to enable comparison with DT characteristics. The first cycle, Studies I and II, was based on the Cross-Disciplinary Workshop for Product Form Design (CDW for PFD) covered in Studies I & II using rapid modelling techniques for bottle form design to facilitate product formation on campus. The second cycle, Study III, was based on the Participatory Workshop for Culturally Oriented Product Design (PW for COPD), and its aim was to facilitate culturally oriented product design in an authentic off-campus field study. RQ1 is addressed in the results section, and RQ2 in the discussion section. A summary of the studies, their focus, and specific research questions is provided in Table 2.

Table 2 The research questions, focus, and specific RQ of each study

Cycles	Study	Focus	Research Questions
Cycle I	Study I (2022)	To analyse problem-solving co-evolution activities/processes in collaborative design	1. How do the groups differ from each other in their DT process? 2. How do the collaborating students carry out the design activities under the task requirements?
	Study II (2023)	To analyse students' design intentions in product design concepts and processes in individual design	What are the main differences in design concepts among students of the three design disciplines (graphic, interior, and product)?
Cycle II	Study III (2024)	To analyse students' design intentions in design concepts and processes	1. What kind of cultural inspiration do the students gain from the local minority culture? 2. How is cultural inspiration mapped onto COPD?

4 RESEARCH DESIGN

Related to research design, the first section introduces the methodological approach of this study based on educational design research (McKenney & Reeves, 2014; 2020), respectively. The second section presents the participants in and settings of the three studies. The third section focuses on data collection and data analysis, while the fourth section addresses ethical considerations.

4.1 Educational design research as the methodological approach

Educational design research (EDR) (McKenney & Reeves, 2014; 2020) is a research methodology, also known as design-based research in education. EDR is a genre of research characterised by the simultaneous development of theoretical insights and practical educational solutions. Its principle is grounded in the idea of conducting ecologically valid studies within authentic learning environments, thereby contributing both to theory building and practical improvements in educational contexts (McKenney & Reeves, 2020). EDR examines educational practices by iteratively designing and implementing educational interventions in a cyclic experimental process. The approach originated in the early 1990s when Brown (1992) introduced the term "design experiment". Over time, it has acquired various names, such as design experiment (Brown, 1992; Bielaczyc, 2013; Collins et al., 2004) and design-based research (Barab, 2006; Sandoval, 2014). McKenney and Reeves (2020) presented the "family tree" of EDR, arguing that these terms are not entirely synonymous. For example, the aim with the original "design experiment" (Brown, 1992; Bielaczyc, 2013; Collins et al., 2004) was to create and test specific learning models, such as reciprocal teaching, within real-world school contexts, exploring both their pragmatic implementation and theoretical implications.

In the education field, EDR is employed as an accepted research paradigm aimed at creating novel learning conditions that are neither common nor well-understood within the learning community (Collins et al., 2004; McKenney & Reeves, 2020). A key characteristic of EDR is its focus on complex, real-world learning settings as opposed to controlled laboratory environments (Brown, 1992). Such emerging learning environments offer space for new perspectives and help refine research questions. Iterative cyclic research can vary in timing and purpose (Ritella & Hakkarainen, 2012). Specifically, the multiple cycles studies were realised by enabling participation in the workshops. Previous EDR-based studies in design education (Lahti, 2008; Seitamaa-Hakkarainen et al., 2017; Riikonen et al., 2020) explored ways to improve how students learn to design.

The methodological approach in this study is based on EDR, consisting of two cycles encompassing sub-studies I, II, and III. All three studies are connected to DT and design practices in product design, conducted in the two design workshop settings. The empirical aim was to enhance both teaching and learning practices in product design while deepening the understanding of DT through practical examination. The motivation for organising the research in this manner is to generate practical results and novel insights into implementing workshop tasks and conditions, namely the CDW for PFD and the PW for COPD, as pedagogical means of supporting learning DT.

A central aim of EDR is the development of theory (McKenney & Reeves, 2014). Theoretical contributions may emerge from educational practice or from within the design research domain (Collins et al., 2004). The theoretical review conducted in this dissertation identifies several domain-specific characteristics of DT relevant to design pedagogy at the university level. EDR was employed in this research to plan, integrate, and evaluate a pedagogical workshop framework for learning DT across two distinct workshop configurations at the YUFE university. These theoretical elements recur across the sub-studies and contribute to the methodological theory of developing DT in varied pedagogical workshop contexts. The workshops were based on project-based working and brief-driven educational practices, while also forming part of the specialised curriculum to be studied and further developed under the product design discipline.

As a senior lecturer in the Department of Product Design at the School of Communication and Design Art (SCDA-YUFE), I held multiple roles within the university's scholarly research programmes, including facilitation, funding, and the resource organisation and management. I was serving as a course teacher, instructor, and researcher in the research projects. The work involved integrating, organising, and managing two-cycle workshops on- and off-campus. During the two cycle workshops, I acted as a course teacher, and I instructed the design students, designed the course objectives and the design tasks, and organised data collections. As the researcher, I was responsible for the practical implementation of the sub-studies and data collection processes in both workshops, undertaking data analysis and thesis writing across the various sub-studies and research projects.

4.2 Participants and settings of the studies

The two design workshops were established as project-based teaching and empirical research at Yunnan University of Finance and Economics (YUFE) in China. These two distinct workshops were structured around the two product-design courses: Rapid Digital Prototyping, and Cultural and Creative Product Design. Specifically, senior design students (third- and fourth-year students) were considered to be optimal participants, as they possessed the necessary design knowledge and practical experience in the design field. To develop T-shape designers, the study assumes that cultivating broad collaborative skills for the cross disciplinary design in university design education, the workshops would deepen secondary areas of expertise by facilitating knowledge cross-over (Nae, 2017). In design

education, design projects become progressively more complicated as studying progresses, facilitating design students for professional practice, mastering embedded knowledge, tools, and skills, and gaining an embodied understanding of the "professional-way-of-being" (Adams et al., 2011).

This encompasses two cycles of the design workshops between 2020 and 2024. The rapid modelling techniques /CDW for PFD was held on campus, while the minority culture context/PW for COPD was conducted off-campus in an authentic field study. The two workshop configurations concentrated on product form with functional features, and on cultural understanding, content, and implications for culturally oriented design. An overview of the workshop settings and participant details is provided in Table 3.

Table 3 The basic information of the three sub-studies in the design workshops

Studies	Design workshops	Participants	Method	Time Duration
Study I	Rapid modelling techniques with collaborative design	9 students in the third year of the undergraduate degree	Design-based experiment with the PFD in CDW and collaborative design	21 hours of learning across 7 weeks
Study II	Rapid modelling techniques with individual design	17 students in the third year of the undergraduate degree	Design-based experiment with the PFD in CDW in individual design	21 hours of learning across 7 weeks
Study III	Minority culture context with individual design	8 students in the fourth year of the undergraduate degree	Design-based experiment with the COPD in PW in individual design	10 continuous days the field study

Study I focused on the CDW for PFD design (collaborative) and rapid modelling techniques applied to the bottle form design task. Student groups were required to develop a detergent bottle form that met user requirements. The study analysed the collaborative design processes of three cross-disciplinary groups (graphic, interior, and product design students). The workshop was part of the selective course, namely Rapid Digital Prototyping, for third-year students and included 21 hours of instruction over seven weeks. Nine student volunteers were divided into three groups based on their disciplines. The entire design process was video-recorded, and the outcomes were collected for analysis.

Study II, also based on the CDW for PFD, used the same condition and task as Study I, but focused on individual design practice. Seventeen third-year students from various design disciplines participated. The data consisted of visual and textual materials, e.g., concept maps, sketches, 3D models, and reflective reports, to investigate individual variations in DT processes.

Study III was based on the PW for COPD and relied on the Maguan Tourism Development design task. Students developed culturally oriented products that aligned with local tourism goals. The workshop was part of a Cultural and Creative Product Design course and was jointly organised by the university and the local government of Maguan County (马关县) in Yunnan Province, China. Eight students voluntarily participated, none

of whom had previously been familiar with the minority culture involved. Field research was conducted through observations and discussions with local people. The data included visual representations, written notes, and recorded explanations of design outcomes.

4.2.1 Cross-Disciplinary Workshop for Product Form Design

For Cycle I (Studies I and II), the course (Rapid Digital Prototyping) addressing rapid modelling techniques comprised 21 hours of training spread over seven weeks. Each session included hands-on assignments (see Table 4). The course focused on computer-aided design (CAD), specifically using SolidWorks. I served as the course teacher and instructor responsible for the course pedagogy and delivering DT-related instruction and structuring the experiment and design tasks.

Table 4 Seven structured sessions for the design task (Studies I & II)

Sessions	Main themes	Case practice
Session 1: Introduction	• Design Intent • File References • The SolidWorks User Interface • Using the Command Manager	Become familiar with the user interface
Session 2: Sketching with SolidWorks	• 2D Sketching • What Are We Going to Sketch? • Sketching Sketch Entities • Basic Sketching	Case practice of sketching entities
Session 3: Sketching with SolidWorks	• Rules That Govern Sketches • Design Intent • Sketch Relations •Sketching on a Planar • Dimensions Extrude	Case practice of bowl and chopsticks
Session 4: Basic Modelling I	• Choosing the Best Profile • Choosing the Sketch Plane • Details of the Part • Boss Features • Revolved	Case practice of writing box
Session 5: Basic Modelling II	• Cut Feature • Filleting • Editing Tools • Linear and Circular Patterns • Hole wizard	Case practice of flower vase
Session 6: Basic Modelling III	• Symmetry • Sketching the Model • Using Model Edges in a Sketch • Creating Trimmed Sketch Geometry	Case practice of scent bottle
Session 7: DT Introduction	• Theory Background • Method & Processes • Design Principles • Commodity Products • Appearance Design	Become familiar with design representations

At the end of the CDW for PFD, cross-disciplinary students either worked collaboratively (Study I) or individually (Study II) to create detergent bottle concepts based on the design brief. In the collaborative design setting, students worked alongside peers from their respective design disciplines. The design brief was formulated to cover some predetermined product requirements of the bottle form.

4.2.2 Participatory Workshop for Minority Culture Context

In the PW for COPD (for the Maguan Tourism Development task), the students from the fourth year investigated cultural elements through field research in the local area. They were tasked with integrating local cultural characteristics into everyday products to generate new cultural product concepts. Study III, part of the cultural product design course called Cultural and Creative Product Design, was co-organised with local authorities. It emphasised on-site inspiration and aimed to translate cultural heritage into product ideas. I served as both the course teacher and instructor. I led the field study and structured the course as a participatory experience, ensuring that each student engaged directly with local people and their culture. The workshop promoted cultural translation through visual, behavioural, and conceptual exploration. Table 5 provides further detail about the participatory practices carried out during the three workshop sessions.

Table 5 The structure of the sessions in the design workshop

Sessions	Involved local participants	Participatory practice
Seven days: Field visits for inspirational resources	• Maguan Area History Museum • Xiao Magu Village • Aboriginal Clothing Street Area • Masa Village • Cultural Core Area of Maguan County	Participatory research in the field: 1. Visiting relevant cultural and tourist sites under the guidance of local experts and engaging in community activities. 2. Writing daily observation notes related to the design task. 3. Conducting interviews, field observations, and online literature research for data collection.
Two days in total: Target product research	Addressing the target product requires a survey in the marketing area. The shopping guides, shopkeepers, and assistants were involved.	Participatory research in the field: 1. Identifying target products and exploring user needs and usage contexts. 2. Considering opportunities for creating culturally oriented product concepts for local tourism.
One day: The implementation of COPD	Students returned to the university to conceptualise and sketch new product ideas inspired by their fieldwork, incorporating local cultural features.	

This workshop used an open-ended task framework. Students were asked to design a culturally oriented product for tourism in Maguan County, but no specific product typology, user, or function was defined in advance.

4.3 Data collection and data analysis

The aim of the study was to analyse the nature of students' DT practices in two product design workshops. Since this dissertation consists of three studies published in peer-reviewed academic journals, every phase of the research process required rigorous analysis. Across the three sub-studies, qualitative data were analysed using both theory-driven and data-driven content analysis (Chi, 1997; Elo & Kyngäs, 2008; Saldaña, 2009). Coding schemes were developed by conducting a preliminary analysis of raw data and aligning the data with the referred theoretical frameworks. A summary of data sources and the focus of the analysis is presented in Table 6.

Table 6 Overview of the data collection, unit of analysis, and content analysis of the three studies

The Studies	Study I	Study II	Study III
Data	Video data Sketches 3D models	Concept maps Sketches 3D models Reflection report	Sketches Survey report Verbal data
Unit of analysis	Design event Video segment Verbal statement	Design event Design concepts Sketches Visualisation 3D model Visualisation	Design event Design concept Sketches Visualisation Verbal statement
Focus of qualitative analysis	Students' DT activities: Focus on identifying problem-solving co-evolution activities in product design concepts	Students' DT activities: focus on identifying the categories of design domain knowledge in product design concepts	Students' DT activities: focus on identifying the categories of cultural levels and cultural elements in product design concepts
Quantitative description	Frequency distribution	Not applicable	Not applicable

Next, an overview of the classification of the qualitative content analyses is outlined.

Study I

The aim with Study I was to explore collaborative DT activities during group-based student design processes. The objective was to understand the differences in DT activities among student groups working on the PFD in CDW, with a particular focus on the co-evolution of problems and solutions. Three groups of students, each composed of members from graphic design, interior design, and product design disciplines, participated in the design exercise. Study I data consisted of video recordings of collaborative sessions, sketches, and 3D models. Although CAD screen recordings were captured, they were excluded from the analysis. In Study I, the emphasis was on group collaboration, with video data analysed through qualitative content analysis to identify DT-relevant segments (Chi, 1997). Emerging insights were synthesised with theory to derive meaningful understandings.

For the video analysis, Ash's (2007) methodological approach was adapted to track the phases of the design process. At the macro level, the video data were segmented into one-minute intervals, within which the occurring design activities were identified. A theory-driven classification scheme was developed, consisting of the following categories: 1. analysing constraints; 2. Ideation; 3. information seeking; 4. Sketching; 5. 3D modelling; and 6. discussion about computer techniques. Based on this categorisation, flowcharts were created for each group to illustrate the progression of design activities throughout the sessions. At the micro level, the analysis focused specifically on the design activities of co-evolution of problem and solution exploration. Co-evolutionary design statements were classified into four categories: 1. Proposal; 2. Definition; 3. evaluation; and 4. reformulation (see Table 7).

Table 7 Categories and excerpts of the co-evolutionary design statements

Categories	Definition	Examples
Proposal	Proposing and explaining a new solution (or new elements of a solution).	...then, is there a little bend? Maybe this half does not need many rounds...
Definition	Explaining, interpreting, and communicating a proposed solution (or proposed elements of a solution) among the design group.	Right, I think so, um...if I used this shape, then I hope... There is a bulge we can grasp using one hand.
Evaluation	Judging a proposed solution regarding the problem expressed.	You gave me a good idea; I think this can increase the volume of the bottle.
Reformulation	Setting a new problem expression or modifying the existing one. The first initial problem expression is considered as a particular reformulation.	...Yeah, have a straight one like that placed in our dormitory.

Study II

The aim with Study II was to explore individual DT activities during students' product design processes. The objective was to examine differences in individual DT activities among students from different disciplines (graphic design, interior design, and product design) within the PFD task in CDW. The data for Study II were collected under the same design conditions as in Study I, using rapid modelling techniques and the bottle form design task. Visual representations from 17 students, including concept maps, sketches, and 3D models, were collected, along with their reflective reports. These data were analysed using ATLAS.ti, qualitative analysis software that facilitated the coding process. The focus of Study II was to investigate students' design representations, employing a data-driven approach to analysis. The coding categories were adapted from Chamorro-Koc et al. (2015), focusing on the types of design concepts. The categories and definitions are presented in Table 8.

Table 8 Coding schemes of the design concepts

Categories	Subcategories	Codes	Definition
Use Context	Situation	ST	A particular situation in which situated issues emerge and lead users to intentional actions with products.
	Intended Use	IU	Indications about the purpose of use of a product or part of the product, which aim at considering effective use in a context.
Knowledge	Description-Based Concept	DBC	Indications about aesthetic aspects such as the placement, shape, pattern, and decoration of a product and its features.
	Principle-Based Concept	PBC	Indications about the relationships between the product's structure and its functionality are usually connected to usability and ergonomics.
Experience	Episodic Data	ED	Reflections of memories of product use that a person has experienced in the past.

Study III

With Study III, the aim was to explore individual DT activities in the setting of COPD in PW. The task was based on the Maguan Tourism Development project. The objective was to investigate how on-site cultural inspiration influenced students' DT and the development of their concepts. Eight product design students participated in this minority cultural heritages participatory study. During the design process, each student was required to produce sketch-based representations to explore and develop new product ideas. Students' design outcomes were documented, and their verbal explanations were audio-recorded and later transcribed. Thus, the data included sketches, text notes, and transcribed verbal explanations of the product concepts. A data-driven analysis approach was used, employing a metaphorical mapping tool (Qin & Ng, 2020) to identify and categorise the cultural levels and symbolic elements embedded in the students' designs. The analytical categories are presented in Table 9.

Table 9 The analytical categories of the three-level structure

Cultural three levels	Cultural elements
Outer level: Corresponding to the tangible category, such as visual symbols, and artefacts, normally in response to graphic elements.	Colour Pattern Materials Form Structure
Intermediate level: Corresponding to behavioural activities that appear in a traditional institution or contemporary lifestyle, which could be selected to build a connection with the outer layer and the inner layer.	Craft Operation Function Expression of customs
Inner level: corresponds to the intangible category, and works as the core value, e.g., from aboriginal beliefs to stir cultural thinking and reflection.	Spiritual values Philosophic thinking Manner of aesthetics

4.4 Ethical considerations

This study adhered to the ethical guidelines provided by the Finnish Advisory Board on Research Integrity (Varantola et al., 2012), which emphasise the principles of responsible conduct of research, including honesty, transparency, respect for research participants, and accountability in data management. Ethical considerations were my concern throughout all the phases: research design, implementation, and dissemination phases.

These empirical studies adopted situated learning to integrate student participants' learning objectives with the research objectives in the various research contexts. As participation aligned with course learning objectives, prior consent for research participation was not required. However, institutional approval was obtained from the design school before each project began. At the start of every project, students were informed about the research, and their voluntary participation was sought. Informed consent forms were distributed in writing and explained verbally. All participants participated voluntarily and were fully informed of each study's situation. The aim of each study was to integrate the design practices into routine design activities. To obtain the reflection of design thinking through practical design activities, each piece of research emphasised engaging students in specific contextual situations within the related design workshop.

Special attention was given to safeguarding participants' privacy, dignity, and confidentiality. To protect anonymity, all personal identifiers were removed from data used in the published findings. Participants' identities were anonymised in all written descriptions, and any images used in dissemination materials or publications were altered (e.g., via blurring) to ensure that individuals could not be recognised.

Assigning the cross-disciplinary design tasks referred to the product design pedagogy with the on-campus courses, which was carefully tailored to align with the prior learning, ensuring relevance and engagement. These varied tasks were integrated with my assistance into their course assignments for both product design students and students from other design disciplines, while avoiding additional workload arising from their participation in the study. Throughout the study, as the researcher, I facilitated and enabled students to gain the product design understanding along with the professional way of being and encouraged open communication and collaboration regarding all aspects related to data collection and participation, while ensuring alignment with the training requirements for developing design capabilities.

In the field study, design students were prepared and facilitated to make observation notes, take photos, and interview residents, during which they could meet various local experts, artisans, museum curators, shopkeepers, and villagers. They also walked across tourist attractions and participated in various events with the local people. These interviews and participatory observations focused on local minority traditions such as rituals, music, and sport. While I facilitated the participants engaging in the field, this also fully considered the potential cultural sensitivities involved in participatory and field-based research, particularly within culturally oriented design contexts. Therefore, ethical reflection was approached as an adaptive process among the participants. Throughout this, I emphasised

the participants' engaging with local communities, learning and understanding other cultures and tacit meanings, and maintaining the utmost respect for the local people and their cultural traditions.

5 RESULTS

This chapter summarises the key findings of each sub-study, while in-depth information about studies can be found in the original publications.

5.1 Study I

Study I focused on collaborative design activities and problem–solution co-evolution among the cross-disciplinary student groups (product design, interior design, and graphic design) engaged in the bottle form design task. A qualitative content analysis was conducted using video recordings and sketches, focusing on the progression of the design processes and the differences within groups. The time spent by the groups on the design processes varied from between 71 and 111 minutes. Detailed analysis focused on the early stages of a design process, that is, sketching a form of the bottle.

The video data collected were based on a video analysis method (Ash, 2007). The analysis showed that the product design students demonstrated the most intensive period of problem–solution co-evolution, as revealed through verbal analysis. The verbal problem-solving activities of the groups were coded into four categories: proposal, definition, evaluation, and reformulation. The duration and frequency of problem-solving statements varied: G-Product (21 min, f = 143), G-Interior (15 min, f = 79), and G-Graphic (40 min, f = 114). These results indicated that the G-product produced the highest number of problem-solving statements within a shorter period. To analyse intensity further, the frequency of problem-solving statements per minute was calculated: G-Product (6.8), G-Interior (5.26), and G-Graphic (2.85). This confirmed that the product design students demonstrated the most intensive problem–solution co-evolution activities. Additionally, the number of statements varied significantly among individual group members.

The findings suggested that the G-product focused more heavily on problem–solving co-evolution activities compared to the other groups. Their creation of new 3D forms was driven by iterative reformulation framing new problems or modifying existing ones. This reflects an ability to evaluate the design context rapidly and project alternative possibilities. In contrast, the other groups emphasised visual aspects of form rather than user-centred concerns, such as usability. Despite these differences, all three groups succeeded in producing a new 3D detergent bottle design that met the PFD task requirements. This underscores the importance of supporting students' understanding of both product form and functionality in cross-disciplinary settings.

Study I also noted the challenge of implementing broader pedagogical objectives within the constraints of a short workshop course. However, from a design education perspective,

students must practise design tasks of various kinds, including well-structured activities that scaffold their learning.

The results from Study I revealed key aspects of the nature of students' problem–solution co-evolution in collaborative settings and how domain-specific knowledge shaped these PFD processes. However, since the analysis relied solely on the recorded product design sessions, it did not examine how students' original disciplinary knowledge influenced their performances.

5.2 Study II

Study II focused on how individual students from different design disciplines (product design, interior design, and graphic design) engage in the bottle form design task. A qualitative case study approach was used to analyse visual and textual design representations, aiming to identify and interpret students' conceptual responses to the product design brief. The data consisted of students' concept maps, sketches, 3D models, and reflective reports. The data were analysed using theory-driven qualitative analysis.

All students successfully developed description-based concepts that met the product design requirements. However, principle-based concepts like those grounded in structural and functional considerations were only evident among the product design students. This suggests that product-domain knowledge significantly influences both design execution and outcomes. To illustrate this distinction, three student cases were compared. Two graphic design students designed the bottle shape using analogies from online searches or personal experience. In contrast, a product design student applied principle-based reasoning to optimise the bottle handle for ergonomics. The student designed a recessed grip for thumb placement and reshaped the handle to ensure even pressure distribution. The bottle cap was also redesigned for ease of opening by increasing friction. This design reflected a clear application of domain-specific knowledge from previous product design learning.

Study II showed that cross-disciplinary students are capable of developing viable product concepts, particularly when focused on 3D form. All students successfully designed new 3D detergent bottles that met PFD task requirements. However, product design students paid more attention to product usability and UCD principles than those from other disciplines. The study also found that combining visual representations with reflective reporting provides a more holistic understanding of student knowledge and DT features.

The results from Study II also revealed the challenges non-product design students face when addressing typical PFD tasks. While such tasks are routine for product design students, they may be unfamiliar to students in other design disciplines. Although this study was limited to document-based materials, it offers valuable insights for design educators. Instructors are encouraged to challenge students' initial assumptions and develop brief structures that guide deeper engagement with user needs and design perspectives.

5.3 Study III

Study III addressed the research questions and focused on how individual students use local cultural experiences as inspiration for culturally oriented products. The analysis employed the metaphorical mapping tool developed by Qin and Ng (2020) to identify cultural levels and elements embedded in the design concepts. The data included visual representations, design notes, and verbal explanations from interview meetings. Each of the eight product design students participated in the field-based design task and produced a total of 13 culturally oriented product concepts. The metaphorical mapping tool helped to analyse cultural inspirations at three levels: outer (tangible), intermediate (behavioural), and inner (intangible or philosophical).

The results showed that most design concepts reflected the outer level, drawing from visual features of local artefacts, sculptures, and clothing. A smaller number addressed the intermediate level, such as daily rituals or cultural practices. Only a few concepts reached the inner level, incorporating spiritual values or philosophical meanings. These findings indicate that students primarily drew upon surface-level cultural features (cf. Qin et al., 2019). However, the tangible elements were effectively transformed into new product contexts. Three representative examples are discussed in the publication, to illustrate how visual, behavioural, and philosophical features were integrated into new product concepts.

Study III also highlighted the importance of the field task and students' direct engagement with local heritage as sources of design inspiration. The metaphorical mapping tool offered a useful framework for identifying cultural influences and generating new meaning during the design process. This approach could also be used in design education to support critical reflection on cultural heritage and product development.

The findings of this study open new perspectives on stimulating creativity in the design of cultural products, and three-level metaphorical mapping could also be used as a pedagogical tool to analyse the sustainable value linking the cultural aspects and the target products. Metaphorical mapping could be used in teaching design to analyse the relationships between various product elements of the cultural heritage. Our results imply that the intermediate or inner level could trigger the realisation of the metaphorical connection, rather than focusing only on the visual or physical aspects of the artefacts. Further, to support students' inner level of design better, the design process could be more scaffolded to support new ways of thinking. These prompts could help students to move beyond the outer level in their DT. Also, developing deeper cultural understandings might require a more extensive period.

6 DISCUSSION

The discussion focuses on the framework of DT (Carlgren et al., 2016) to identify, compare, and interpret the essential DT features that emerge in the two comparative workshops. Studies I and II are based on the cross-disciplinary of CDW for PFD on campus, with rapid modelling techniques for the bottle form design. Study III is based on the participatory of PW for COPD off-campus in an authentic field study. In particular, Study III pays close attention to the role of institutional collaboration between academic actors (YUFE) and local Maguan government bodies (马关县旅游局) in creating meaningful field contexts. This setting exemplifies how diversity and cultural complexity inform the DT process in the participatory workshop.

The discussion highlights the differences arising from disciplinary backgrounds and the nature of the design tasks. Five comparative pairs are used to frame the discussion:

1. User focus: Designer-as-user approach vs. Direct user involvement
2. Problem framing: Well-structured vs. Open-ended tasks
3. Visualisation practices: Talking sketches vs. Thinking sketches
4. Experimentation: Product form design vs. Culturally oriented design
5. Diversity in participation: Collaborative group design vs. Individual design

User focus: Designer-as-user approach vs. Direct user involvement

In DT pedagogy, the nature of user involvement significantly shapes the orientation and depth of the design process. As noted by Liedtka (2015), user-centredness is not merely a methodological concern but is also a philosophical orientation that prioritises empathy, user participation, and contextual sensitivity. User needs can be addressed by product design solutions, either through direct participation or indirect involvement, highlighting the importance of understanding the emotional, cultural, and design contexts.

In the cross-disciplinary workshop for designing product forms (Studies I and II), students are not required to interact with actual users. Instead, they work from predefined briefs (e.g., the bottle form design task) and rely on simulated user personas or general assumptions about potential end-users. This form of user engagement aligns with what Zhang and Dong (2016) term the “designer-as-user” model in which the designer’s imagination and personal experience substitute for real user input. Product design students tend to use relatable user scenarios, such as referencing “family members” as representative target users, indicating an inclusive consideration of all members within a household (i.e.,

the home context). This suggests that student designers frame their scenarios around practical use situations when diverse users within a family might interact with the product design. In contrast, interior and graphic design students often focus on more narrowly defined user segments, such as those concerning age and gender. While such a targeted perspective may yield inclusively customised solutions, a design-for-all mindset encourages students to consider a broader range of user experiences. Conversely, designing for a single user segment may overlook the diverse needs of other potential users and limit the broader applicability of design solutions.

By contrast, the participatory workshop for culturally oriented products (Study III) integrates direct user involvement through participatory ethnographic fieldwork. Students engage with users through interviews, participant observation, and collaboration with local communities (in Maguan County). This approach enables a much richer understanding of users as cultural actors embedded in social, historical, and symbolic systems. Students attend local festivals, visit artisan workshops, and participate in cultural rituals, which allow them to capture not only explicit user needs but also tacit knowledge, such as material culture, narrative identity, and emotional resonance. This leads to product designs being grounded in symbolic references (e.g., products inspired by the Masa horse) and embedded in local aesthetic traditions and daily routines. This level of engagement transforms users into co-creators of meaning, shifting the role of designers from a detached expert to a reflective facilitator and interpreter. In this model, the user becomes not just a data source, but also a carrier of heritage, values, and co-authored futures.

Problem framing: Well-structured vs. Open-ended tasks

DT, as proposed by theory scholars such as Simon (1973) and Schön (1983), involves continuous reframing of problems in response to emerging insights and contextual feedback. The design task structure plays a crucial role in shaping this reflective, iterative process (Kapkın & Joines, 2021).

The bottle form design task in the cross-disciplinary design setting is technically constrained and highly defined: redesigning a bottle for increased capacity without compromising usability or formal aesthetics. This well-structured problem type provides clear requirements and expectations:

- Usability, ergonomics, and production feasibility are foregrounded.
- Design iterations focus on ideation, form operations, CAD modelling, and prototyping logic.

The problem-solving involves design concept reasoning, product sensitivity, and precise modelling.

The participatory workshop uses an open-ended task with which students are asked to design a culturally oriented product for tourism development in Maguan County, but no

specific product typology, user, or function is defined in advance. The design problem must be discovered, not simply solved. This requires:

- Deep contextual immersive activities through fieldwork and dialogue
- Empathetic understanding of local cultural dynamics
- Negotiation of competing values: tradition vs. innovation, local vs. tourist perspectives

Here, problem framing is dynamic, shaped by real-time interactions with the various communities. Students do not just define a problem, they co-construct it with users and stakeholders. This process fosters sensitive DT skills and design resilience—abilities crucial for addressing complex, ambiguous design challenges in real-world settings.

Visualisation: Talking sketches vs. Thinking sketches

Sketching and visual prototyping are integral to how designers externalise cognition, collaborate, and test hypotheses (Goldschmidt, 1991; Ferguson, 1992; Akoury, 2020). This discussion distinguishes between talking sketches and thinking sketches, based on their function and social context.

In the cross-disciplinary setting, sketching is highly social and iterative. Students create talking sketches, drawings developed in conversation with team members, used to develop partial ideas, provoke dialogue, and move from ideation to sketching in a shared visual language. Group sketching sessions involve real-time co-creation. For instance, graphic design students use sketching trends for visual representations, and product design students use sketching trends to build usability and functional solutions. Sketches evolve quickly into CAD models, enabling iterative development. These sketches function as boundary objects (Star & Griesemer, 1989), tools that bridge cognitive gaps between group members in collaboration.

In contrast, sketching is an individual reflective activity in the participatory design. Students use thinking sketches to translate cultural insights into symbolic design ideas. They reflect on material traditions, cultural meanings, and symbolic associations and formulate culturally sensitive narratives before design resolution. These sketches are often created alone, in field notebooks, or through participatory observation. They are deeply interpretive, capturing not only what students see but also what they understand. Later, these sketches are used in feedback sessions, making them both internal cognitive artefacts and external communication tools. However, their primary purpose remains culturally interpretive reflection, bridging cultural memory with modern design meaning.

Experimentation: Product form design vs. Culturally oriented design

Experimentation is a core principle in DT and a defining pedagogical element in design education. As emphasised by Carlgren et al. (2016), it is one of the five essential

characteristics, representing a mindset that supports iterative exploration through key phases of a process. In educational contexts, experimentation not only enables students to refine their design outcomes but also reshapes how problems are framed and reformulated within situated learning environments.

In the cross-disciplinary setting, experimentation is structured within a workshop-based, cross-disciplinary setting focused on bottle form design. The objective is to develop product appearance through iterative activities such as ideation, sketching, and CAD-based modelling. It centres on product form development, emphasising end-user needs, aesthetics, and ergonomics within a structured task brief. Over seven weeks, students from product, interior, and graphic design disciplines participate in 21 hours of structured instruction in SolidWorks-based rapid modelling, guided by DT principles. Students explore formal variations by manipulating product form and function, adjusting design proportions, and iterating between hand sketches and digital 3D models. Group members engage in peer critique, and instructor feedback is systematically integrated into the problem-solving cycles. The workshop includes both collaborative and individual design modes. In group sessions, students test ideas collectively and refine features in response to visual and functional criteria, promoting mutual learning and collective decision-making. The learning conditions emphasise iterative making, visual experimentation, and real-time feedback. By integrating various design representation sections as a learning pathway, these structured, form-centred conditions, provide an experimental pedagogical platform.

In participatory design, this addresses the product metaphor design, translating cultural symbols, rituals, and local artefacts into new product concepts that reflect collective cultural identity. Supported by academic and local cultural authorities, students use the three-phase COPD methods, field research exploration, and visual metaphor mapping to explore community values and reinterpret them through design. The design process emerges through cultural immersion and meaning making in the field. Students develop product concepts iteratively in response to conversations with local actors, such as artisans, museum curators, and shopkeepers, allowing ideas to evolve through situated understanding and metaphorical reinterpretation. Although students work individually, they do so within a participatory framework. External perspectives from multiple stakeholders are continuously integrated into the design process, resulting in evolving metaphoric interpretations of local culture. Local villages become living classrooms, where rituals, materials, and everyday interactions provide unpredictable yet meaningful feedback. These conditions demand cultural, empathetic, and adaptive DT. Unlike the tool-structured learning in the cross-disciplinary workshop, this setting requires students to navigate ambiguity and interpret symbolic content, emphasising conceptual transformation rather than CAD-based technical resolution.

The paired comparison reveals two distinct models, shaped by differing design foci and learning conditions. In the cross-disciplinary workshop, experimentation is centred on the manipulation of product form within a controlled studio environment and a well-defined task. Students engage collaboratively or individually in iterative processes that focus on aesthetics, ergonomics, and functional design through tools such as sketching and SolidWorks. In contrast, the participatory workshop frames experimentation within a

participatory, real-world context defined by ill-structured problems. Students work individually yet in dialogue with local people to reinterpret cultural narratives through product metaphor design. Here, experimentation shifts away from the refinement of physical form toward the exploration of local cultural meaning.

Diversity: Collaborative group design vs. Individual design

Participation is central to both DT and contemporary pedagogy, particularly in fostering intercultural understanding, inclusion, and co-creation (Manzini, 2016; Van Oorschot et al., 2022). This study examines how two types of participation support diverse DT learning trajectories.

Students work in cross-disciplinary groups, combining product, interior, and graphic design students. This structure promotes 1. cognitive diversity: students bring different skills based on their original major, habits of mindset, and aesthetic sensibilities, and 2. negotiated collaboration: design solutions are formed through compromise, visual dialogue, and collective testing. However, this model also reveals asymmetries of participation. The product design students, due to stronger product modelling skills, often dominate technical decisions. Interior and graphic design students, while visually creative, sometimes lack functional reflections. Despite equal access to tools, disciplinary knowledge (i.e., skills and techniques) structures group dynamics. Still, the collaborative model facilitates multivocal ideas, which enhance the creative richness of the design outcomes.

Participation with the participatory setting is individual but embedded in a community-concerned framework. Students develop their design proposals while engaging with local stakeholders (e.g., artisans, craftspeople, and community members). Participation here is 1. reciprocal: students contribute design skills, while community members share local cultural knowledge; 2. situated learning: students immerse themselves in lived experiences of the local context; and 3. ethnic-cultural: students reflect on their responsibility as external designers interpreting cultural identity. Rather than co-designing with peers, students co-design with culture bearers. This type of participation demands humility, self-awareness, and a deep commitment to representational accuracy and cultural respect (Mavri et al., 2021).

In summary, the diversity of the cross-disciplinary workshop emerges from a cross-disciplinary teaching setting—ideal for exploring design performance and group work. The diversity of the participatory workshop arises from cultural immersion—ideal for ethical, localised, and cultural design learning.

The five comparative discussions illustrate that DT is not a static method but a situated, evolving pedagogical practice. Each setting structure offers different affordances for learning. The cross-disciplinary approach supports design skill development, collaboration, and procedural mastery, whereas the participatory approach fosters cultural awareness, empathy, and contextual engagement. The user perspective, problem framing, visualisation, experimentation, and diversity dimensions form a multi-layered framework for teaching DT in product design education. Thoughtfully balancing these aspects can cultivate inclusive, critical, and innovative designers for an increasingly pluralistic world.

7 RESEARCH IMPLICATIONS

This section distils the key insights generated by the two-cycle, design workshop-based study and explains those matters beyond the present study. First, it positions the findings within current DT scholarship, showing how students' practices in product-design workshops enrich debates on co-evolution, inclusivity, and cultural sensitivity. Second, it translates those theoretical advances into pedagogical guidance for design educators who wish to strengthen empathy, collaboration, and user-centred problem-solving in design courses.

7.1. Theoretical contributions to design thinking research

This study aims to understand the nature of students' DT practices within product-design workshops. Accordingly, the focus is placed on three key theoretical aspects of DT activity in product concept design: domain-knowledge-oriented co-evolutionary activities in collaboration, broadly inclusive design intentions in product form design (PFD), and culturally sensitive activity in culturally oriented product design (COPD).

7.1.1 Domain-knowledge-oriented co-evolutionary activities in collaboration

Product form design often rewards eye-catching form, leaving ergonomics, usability, and other domain considerations under exploration. This visual-focused perspective weakens creative concept quality because successful products emerge from the co-evolutionary problem-solving in dialogue with domain knowledge (Maher & Poon, 1996; Maher & Tang, 2003; Dorst & Cross, 2001; Lonchampt et al., 2004), such as human factors, user surveys, ergonomics, and business goals. Building on the co-evolution literature (Lotz et al., 2015; Ball & Christensen, 2019; Dorst, 2019), focus on how targeted, domain-specific activities catalyse richer problem-solution cycles in collaborative design in the field.

Co-evolutionary design activity hinges on effective reformulation (Dorst & Cross, 2001; Lonchampt et al., 2004), as designers sketch a tentative form, new functional or user constraints surface, prompting a fresh problem frame (Self, 2017). Without a knowledge base, e.g., anthropometric data, usability analyses, lack the criteria to recognise richer reformulations. The present study showed that the product design students with the ergonomic-specific knowledge and used user-perspective test rigs cycled through more problem–solution exploration loops and produced significantly more viable concepts than the other groups guided solely by aesthetic judgements.

Crilly et al. (2009) note that form must simultaneously attract attention, signal product type, user research, and convey conceptual function. Purely visual exploration addresses the first two aims, but stalls at the latter pair. In the present study, the product design students trained in the user-centred approach ran quick usability walk-throughs between iterative problem-solving loops; each walk-through surfaced latent use constraints in the context of use that facilitated reforming the conceptual product appearance and geometry. These cycles confirm that evaluation activities are domain-knowledge dependent (Ahmed et al., 2003; Cross, 2004) and should be enhanced, if without domain inputs, the feedback loop collapses into superficial visual aspects. In prior studies, co-design teams through mixed-discipline groups let product students share human-factors tools while peers contribute branding or spatial insights, mirroring real-world DT practice (de Vere et al., 2010; Fleischmann, 2022).

The findings extend the co-evolution theory by demonstrating its dependency on domain-specific knowledge in the product design field, in which richer domain knowledge widens the adjacent knowledge possible during reformulation activities in collaboration. Pedagogically, teaching would integrate key domain modules, such as user-centred perspective, into concept design learning and formalise cross-disciplinary critique sessions. Design practice shifts attention from surface aesthetics to the deeper, knowledge-driven dynamics that widen product design concepts.

7.1.2 Broadly inclusive design intentions in designing product forms

In cross-disciplinary design education, a recurring challenge in product concept design is the prevalence of design fixation (Nicholl & McLellan, 2007; De Vere et al., 2010; Liedtka, 2015), particularly among students without formal training in product design. These students often rely on prior learning experiences, superficial inspirations, or examples from adjacent fields, rather than engaging with domain-specific knowledge and broadly inclusive DT activities (Sanders & Stappers, 2008; Andreasen et al., 2015). As a result, their outputs tend to reflect simplified and stereotypical assumptions about users, which narrows both the relevance and usability of their design solutions.

Nicholl and McLellan (2007) argue that routine product analysis, while intended to cultivate critical insight, can inadvertently encourage the reproduction of familiar design tropes. Visual and functional features of exemplar products are replicated without meaningful reinterpretation, thus reinforcing design fixation. This effect is even more pronounced in cross-disciplinary environments, where the lack of grounded exposure to user diversity leads students to default to limited user typologies. These limitations stem from cognitive bias, as described by Liedtka (2015), where constrained mental models and normative expectations prevent students from fully exploring innovative or inclusive possibilities.

This divergence underscores the importance of embedding UCD principles and reflective thinking activities throughout all design stages, particularly important for students outside product design. Furthermore, the study highlights how visual examples, commonly

used as stimuli in design workshops, may inadvertently constrain ideation by solidifying default assumptions and inhibiting exploratory thought. This finding aligns with Sawyer's (2012; 2018) research on creativity, which emphasises the need for iterative, critical, and collaborative approaches in developing novel ideas. According to Andreasen et al. (2015), conceptual design relies heavily on framing and reframing user problems and potential solutions, a process that is short-circuited when students are overly dependent on static references or surface-level features.

To mitigate these challenges and promote broadly inclusive design intentions, the following pedagogical strategies are proposed:

1. Reframe product analysis as a speculative, critical practice. Rather than replicating, students should be encouraged to question the rationale behind existing design features, generating alternative forms based on diverse user segments. This approach mirrors the emphasis on making and reflection in co-design practices (Sanders & Stappers, 2008; 2014).
2. Shift focus on user needs and real-world contexts. Embedding tools such as user scenario development or contextual inquiry (Crilly et al., 2009) can move students beyond aesthetic appreciation to engage in an empathetic understanding of lived user experiences.
3. Embedded iterative and reflective learning processes. Exercises such as design journaling or self-assessment can reveal hidden biases and assumptions about users. Sawyer (2018) argues that creative learning is strengthened when students continuously reflect on their evolving ideas and assumptions.

In sum, promoting inclusive design thinking in product form development requires more than teaching idea generation techniques. It demands a systematic redesign of pedagogical environments to support reflective, critical, and user-aware thinking. Through strategies that directly address fixation, educators can foster richer and more equitable design practices, equipping designers to develop products that serve a wider and more diverse user base (Liedtka, 2015; Sanders & Stappers, 2008).

7.1.3 Culturally sensitive activity in designing cultural products

Culturally oriented product design is increasingly recognised as a promising way to embed cultural identity in product innovation, particularly when design intersects with heritage preservation and social meaning (Luo & Dong, 2017; Chai et al., 2018; Moalosi et al., 2010; Qin & Ng, 2020). Yet the design process still faces three persistent contextual challenges: 1. reliance on second-hand cultural sources; 2. limited authentic cultural experience; and 3. cultural appropriation risk. Culturally sensitive activities through participation provide an effective means of addressing these challenges while broadening the research implications. Such activities not only enrich the design process; they transform field research into embodied insight, converting cultural knowledge into authentic, lived understanding through participatory engagement (Manzini, 2016; Van Oorschot et al., 2022; Roque, 2024).

Although guidelines, literature, and museum collections offer an accessible starting point for design (Hsu et al., 2011; Luo & Dong, 2017), they rarely support the deep contextual transformation of knowledge required in COPD. These sources privilege representational over experiential learning, limiting designers' ability to translate abstract cultural insights into contextually situated product concepts (Moalosi et al., 2010). By contrast, culturally sensitive activities, such as immersion in festivals, craft workshops, and social rituals, allow designers to engage directly with both tangible and intangible cultural dimensions (Chai et al., 2018; Qin & Ng, 2020). Participation theory (Van Oorschot et al., 2022) underscores that meaning making arises through embedded, embodied activity. In this study, through a participatory approach, such active engagement facilitated the shift from passive learning to situated understanding, echoing the emphasis by Roque et al. (2024) on community-based participant observation as a route to cultural sensitivity and situated knowledge. Participatory engagement therefore strengthens designers' capacity to internalise cultural symbolism and reflect it authentically in the social aspect.

Authenticity in cultural product design hinges not on the superficial replication of symbols but on integrating cultural design practices, values, and lived experiences into the design process (Qin & Ng, 2020). Local communities provide rich contexts that grant access to deep cultural, e.g., rituals, behavioural patterns, and environmental relationships (Moalosi et al., 2010). In the present study, culturally sensitive activities such as participatory observation, in-depth interviews, artefact-handling, and field documentation with recording technology, e.g., field notes, photography, and video recording, enabled engagement with the cultural resonance embedded in local practices. This embedded engagement supports collaborative meaning making. The emotional and cultural awareness fostered through such activities aligns with Manzini's (2016) notion of dialogic design, in which mutual respect and shared understanding become the foundation for design innovation. Consequently, culturally oriented product design can function as a vehicle for both knowledge co-creation and ethically grounded cultural engagement.

Cultural appropriation, through which cultural elements are extracted and commodified without proper contextual acknowledgement, remains one of the more pressing ethical concerns about culturally oriented products. Superficial appropriation is especially likely when designers have only limited exposure to the cultures in question. However, culturally sensitive participatory activities establish structures for shared authorship and dialogic feedback. In this study, a collaboration between Yunnan University of Finance and Economics (YUFE) and the local authorities of Maguan County exemplified such a co-design approach. Informants, including artisans, curators, and tour leaders, served as boundary objects mediating between local cultural logics and design interpretation (Suib et al., 2020). This participatory validation effect accords with the findings of Mavri et al. (2020), who highlight the value of iterative co-creation and feedback between designers and community stakeholders. By fostering shared ownership of outcomes, these approaches help mitigate misrepresentation and promote ethical, respectful interventions.

7.2 Practical implications for enhancing design thinking

The present study emphasises how educators can use product concept design workshops to support design students' DT learning, helping young designers develop product concepts through a structured approach. Drawing on insights from the two-cycle workshop research, the study proposes pedagogical directions for design educators, employing a user-centred approach that follows two key dimensions: enhancing empathy and fostering social interactions in collaboration.

7.2.1 Enhance empathy

A user-centred approach is fundamental to DT, placing human needs and experiences at the core of a design process. In design education, this mindset should be cultivated through structured methods that teach students how to analyse, identify, and interpret user involvements, e.g., requirements and experiences. This study emphasises the importance of empathy-based learning in product-design workshops and draws on the Hasso Plattner Institute's DT model to deepen this approach. It refines the empathise phase in DT by dividing it into two clear stages: understand and observe. In the understand stage, students seek and collect background information, e.g., demographics, theories, and assumptions, to define the design focus and plan user research. In the observe stage, they conduct fieldwork through interviews, observation, or immersion to gain real insights into user behaviours and needs. This two-step approach helps students move beyond surface-level assumptions, supports deeper empathy, and provides educators with a practical structure for guiding UCD design.

This practical engagement enables students to detect both explicit and latent needs—insights often overlooked by quantitative approaches—and builds the foundation for authentic, user-informed problem framing. Empathy is not only a research phase but also skills of creativity and critical thinking. In design workshops, design briefs should encourage early and continued user exploration, through direct contact with stakeholders or indirect research methods. This supports the development of design intentions grounded in real needs. Chamorro-Koc et al. (2015) suggest that students should engage in diverse representational practices, such as sketching, prototyping, and modelling, to visualise and test their reflective concepts. The present study found that the guided tasks helped students to practise their ideas effectively and iteratively, in line with the Hasso Plattner Institute's iterative and non-linear approach to DT practice. Moreover, Hsieh et al. (2021) emphasise that sequencing design stages into coherent, interconnected parts enhances both DT learning and students' form-development capabilities. The workshop structure should therefore reflect a looped process, wherein students revisit the understand-and-observe stages to refine their ideas and deepen empathy through iteration.

In addition to focusing on users, students must also learn how various stakeholders shape and constrain product form. These constraints may include brand identity, production capabilities, visual-language conventions, ergonomic requirements, and even cultural

expectations (Crilly et al., 2009; Harvey & Stanton, 2013). Understanding and balancing these demands is an essential part of DT. By reconciling stakeholder input through visual representation and prototyping, students begin to develop more responsible and contextually relevant design solutions.

Using the empathise perspective in the early stage as a foundation, educators can structure UCD design workshops that help students build meaningful engagement with users from the outset. Through the understand and observe stages, students not only learn to research and empathise with users but also integrate their insights into design decisions building critical DT capabilities, creativity, and user sensitivity across various design disciplines.

7.2.2 Enhance social interactions in collaboration

It is crucial to consider how to facilitate collaboration among cross-disciplinary design students in achieving business-oriented goals. This study found that collaborative design approaches can advance DT practice more effectively, as reflected in both the cross-disciplinary workshop and the participatory workshop. The findings indicated that integrating participatory design with culturally oriented product design supports the sustainable development of local cultural assets, such as cultural symbols, crafts, and traditions. This involves developing new product concepts or adapting solutions to suit either the original or a newly contextualised situation. The "authenticity" experience is a crucial factor in stimulating creativity within the field. Furthermore, creative outcomes should be both novel and contextually relevant (Mavri et al., 2020). Achieving this may involve new product concepts or the reframing of contextual concerns of existing solutions to fit either original or newly defined situations. For student designers, this requires professional consideration of local stakeholders' needs and cultural contexts, as well as more frequent and detailed interactions with those related stakeholders. Therefore, the cross-disciplinary learning environment should facilitate detailed engagement in design practice within authentic, real-world contexts.

Social interactions act as validation mechanisms, both among student participants on-campus and with local participants off-campus. These collaborative relationships foster ideation and knowledge transfer. Students' personal experiences from fieldwork can be transformed into new knowledge domains. According to Nonaka (1994), socialisation involves sharing tacit knowledge through interpersonal interactions, while externalisation refers to articulating that knowledge into explicit forms accessible to others. This study highlights the significance of dialogue and reflection among participants, which facilitates communication between academic and community contexts. In the participatory workshop, the involvement of local instructors, museum guides, and artisans played a vital role in constructing shared knowledge about local heritage. Successful knowledge externalisation requires constructive collaboration and mutual trust, conditions necessary for sharing original, experience-based insights that are fundamental to tacit knowledge (Nonaka, 1994). In the cross-disciplinary workshop, co-evolutionary design activities demonstrated an

effective way to address well-defined problems, allowing for greater flexibility in problem-solving. These findings indicated that social influence among design-group members supports the simultaneous exploration of iterative problem-solving.

In summary, these implementations are rooted in a product-design and workshop-based pedagogy and can contribute effectively to the development of students' DT learning and expertise. The core idea of DT practice is for design students to adopt user-centred problem-solving in seeking ideal solutions. DT is a design activity centred on exploring possibilities through logic, imagination, innovation, and iterative reasoning.

7.3 Limitations of the study

This study has several limitations. It was conducted through two workshops and three empirical studies organised within the context of Chinese university-level design education. The main limitations concern the research context and participants, data-collection methods, and data-analysis procedures. The following section discusses in detail how these limitations may affect trustworthiness.

First, differences in the research context and participant profiles may have influenced the research. The present design workshops are based on the specific context of Chinese art and design education. At Yunnan University of Finance and Economics, for example, teaching still follows a form-focused tradition, design courses emphasise form-oriented learning and design composition theories inherited from the western modern design field, stressing vertical, discipline-specific learning and giving little weight to cross-disciplinary pedagogy. This contrasts with Nordic schools such as Aalto University in Finland, where curricula tend to integrate design, engineering, and business.

All students in the two workshops and three studies are later-level undergraduates (third and fourth years). Having studied for several years within their design majors, they had already accumulated considerable discipline-specific skills and domain knowledge. When they tackled a product-form task, their prior experience produced clear differences in design outcomes. While these empirical results highlight the value of studying a form-centred, single-discipline learning model, they also mean that the present findings may differ from those that would emerge under other pedagogical approaches to art and design education.

Second, the cross-disciplinary studies focused chiefly on rapid modelling techniques. Participants' personal interest in CAD technology and their prior exposure to relevant CAD tools therefore matter, as some students possessed technical skills and motivations that could exceed those of peers from the same discipline. Differences in learning background, interests, experience, and knowledge undoubtedly affected team collaboration and design results, yet these factors were not examined. Although each group consisted of senior students (which may have dampened extreme disparities), pre-project interviews or questionnaires might be used to reveal how individual factors played out. However, because collaborative design is open-ended and unpredictable, it is difficult to decide in advance which dimensions of students' personal qualities should be examined. Any such data would

inevitably be partial, and interpretations based on incomplete information could weaken rather than strengthen the results.

Third, the validity of the findings rests on careful data collection and iterative interpretation. As a primarily qualitative research method, the sample size may restrict the findings. While the study followed a single group of participants through specific design activities and drew qualitative conclusions, complementary quantitative research is still needed. Each empirical phase therefore requires a rigorous review of student outputs, taking account of the fact that senior students already possess the skills to complete practical design tasks independently.

Multiple data sources, e.g., mind maps, sketches, physical models, audio recordings, video recordings, design reports, and retrospective reports, were collected, with at least three types for each sub-study, providing corroborative evidence across data analyses. Crucially, pairing participants' design outcomes with their retrospective reports helped build a holistic view of how personal experience and knowledge inform product-design activities. Analytical procedures were highly agreed upon by wider research collaboration with other researchers, and diverse joint approaches to data analysis and evaluation were applied throughout each study, ensuring the reasonableness and consistency of the analytical results.

Fourth, in this study, the analysis was conducted by comparing the experiences of product design students, specifically through using various product design tasks as a means of data collection. Some analyses also pointed to product design issues; thus, the study placed particular emphasis on the related competencies. However, interior design students and graphic design students also possess discipline-specific strengths, yet these strengths were not addressed more. Because of these, it represents only one of the design professions involved in the comparison. As a limitation of the study, it is proposed that such collaborative activities be considered as a selective course offering for design students from various disciplines. The intention is that collaborative design should be conducted in mixed, cross-disciplinary groups rather than within separate teams.

7.4 Suggestions

The design workshops also reflect broader shifts in the design industry, within which collaborative, project-based, and transdisciplinary practices are becoming the norm (Rodgers & Bremner, 2019; Zhang et al., 2015; Kim & Yoon, 2012). Today's designers must operate within fluid teams that span marketing, technology, and service systems. Educators can simulate this complexity through design workshop configurations that replicate real-world design ecosystems (Fleischmann, 2022). For instance, based on SDRC-YUFE (Sustainable Design Research Centre-YUFE), some service design projects acquired establishment as a synthesised form at YUFE, whereby I and the SDRC team have integrated design pedagogy and research works across multiple lines while driving co-innovation goals among design projects and scholarly research programmes for sustainable design goals. Connected to my related design study (Yang, 2024a), a service design workshop was conducted within a commercial context focused on the flower marketing

industry, namely the renowned Dounan Flower Market (斗南花卉市场) in Kunming (昆明). The workshop developed a product-service system (PSS) for an online-to-offline flower-selling platform, referred to as the FFRDSS PSS concept. This system utilised a TRIZ-based method (Ekmekci & İstanbul, 2019; Zhang et al., 2003a; Zhang et al., 2003b; Yang, 2024a; 2025) to integrate diverse user needs, logistics, and platform functions, addressing the multifaceted nature of stakeholder coordination and service delivery. This case shows how the service design workshop configuration can confront the complexity of contemporary service ecosystems.

The suggested study introduces a service design team that could activate systems thinking and collaborative problem-solving across multiple touchpoints, ranging from customer interaction to supply chain optimisation, providing a comprehensive service design solution (Fernandes et al., 2020) and producing sustainable, innovative, and culturally sensitive products (Yang, 2024b). In follow-up research, the author proposes a "SmaSer (Smart Service)" analytical framework (Yang, 2025), which applies principles of explainable AI (Gunning & Aha, 2019; Gunning et al., 2019) and the service-blueprint method to guide product-service innovation in smart environments. This framework further exemplifies how design workshops can be structured around advanced methodological tools, such as service blueprints, to encourage systematic coherence and stakeholder collaboration in service processes. DT has also been applied to design strategy and marketing, supporting the development of internal social networks and informal structures for innovation (Kim & Yoon, 2012).

Moreover, the service design setting does more than simulate service delivery; it also enhances social interaction and knowledge creation. As demonstrated in both collaborative and participatory workshop settings (e.g., cross-disciplinary workshop and participatory workshop), cross-disciplinary students benefit from dialogic and reflective engagement with others, transforming field-based observations into shared knowledge through processes of socialisation and externalisation (Nonaka, 1994). These social interactions function as validation mechanisms, both within student groups and with external community members. This not only fosters creativity and ideation but also supports the co-evolution of problem-and-solution spaces in response to ill-defined or emerging needs.

Collectively, these findings underscore that the product–service design workshops are most effective when grounded in collaborative, socially interactive, and user-centred learning environments. Educators can harness this pedagogical model to develop students' DT capabilities, particularly their ability to navigate complexity, balance user needs with system-level constraints, and engage meaningfully with diverse stakeholder groups. The outputs of such workshops reflect not only ideal DT pedagogical practices but also the development of emerging expert capability in real-world product–service innovation.

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ISBN 978-952-84-1568-8 (Paperback)
ISBN 978-952-84-1567-1 (PDF)
ISSN 2954-2898 (Paperback)
ISSN 2954-2952 (PDF)
Unigrafia, Helsinki
Printed in PunaMusta, Joensuu 2025